

Aditi Rao

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Aspiring AI/ML Software Engineer | AI major@ CMU | AI/ML software Intern at HG Insights & Zessta | Seeking Full-time role starting May 2026

EDUCATION

Carnegie Mellon University (CMU), School of Computer Science, Pittsburgh, PA

May 2026

BS in Artificial Intelligence | Minor: Product Management | GPA: 3.56/4.0

Honors: Dean's List (Fall 2024)

RELEVANT COURSEWORK

Machine Learning (ML), Autonomous AI Agents, Natural Language Processing (NLP), Design for AI Products, Generative AI, Speech AI, Ethics Policy computing, AI Representation & problem solving, Computer Systems, Parallel & Sequential Data Structures & Algorithms, Imperative & Functional Programming, Probability Theory, Modern Regression, Matrices, Product Management, Strategy for High-Tech Products, Designing Human-Centered Software, Human AI Interaction, Advertising & Marketing Communication, Marketing & Microeconomics

SKILLS

- **Programming Languages:** Python, C, OCaml,
- **Frameworks/Tools:** PyTorch, TensorFlow, Hugging Face, NumPy, Pandas, ROS, OpenCV, Google OR-Tools, Python transitions, GitHub, Linux, Matplotlib, LaTeX, Tkinter
- **Core Expertise:** Machine Learning, Deep Learning, NLP, LLMs, Computer Vision, Reinforcement Learning, Neural Networks, Data Structures, Algorithms, OOP
- **Other:** Agile/Scrum, Software Development Life Cycle, cross-disciplinary teamwork, Leadership skills, strong communication, team player

EXPERIENCE

AI Software Engineer intern – HG Insights

May-June 2025

- Built an AI driven product-intent match search system in Python using vector embeddings and semantic search **replacing legacy tool**; improved **accuracy by 45%** and scalability for enterprise use - by more than 25+ employees.
- Redesigned search features increasing input **flexibility by 50%** and boosting enterprise **user satisfaction**.
- Designed and evaluated multiple AI architecture approaches – leveraging multi-agent system, Large Language Models (LLMs), ReAct frameworks, and keyword extraction – to build a smarter, faster intent-matching solution.
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AI Software Engineer Intern - Zessta

May - July 2024

- Developed transformer-based **NLP model** (Hugging Face) to **auto-generate Python debug** questions, increasing question diversity by 35%
- Optimized model output (temperature, top-k, top-p tuning) delivering 2 **production-ready ML models** integrated into platform
- Explored watermarking, perplexity, and burstiness techniques to enhance content uniqueness and detection accuracy.

Software Intern - mPowero Elearning

May - Nov 2020

- Partnered with PM team to gather/prioritize feature requests; drove **delivery of 8 new product features** adopted by 2k+ users
- Designed and executed 50+ test cases across mobile/desktop; improved app **performance by 7%**

PROJECTS

Autonomous AI Agent – Machine Learning, Python, ROS, Computer Vision

Oct - Dec 2024

- Developed an autonomous agent using ROS and Finite State Machines to optimize plant growth in controlled environments through real-time data processing. Collaborated within a three-member team, driving end-to-end development from design to testing.
- Applied techniques of Explanations ML, and Computer Vision to detect and fix faults, boosting system reliability by 40%.
- Engineered workflows with Python's transition library, implementing Scheduling, Deployment, Execution Monitoring, and Testing for robust automation.

Systems Programming Projects (C):

Oct - Dec 2024

- Built multi-threaded proxy server, Unix-like shell with job control, custom memory allocator, and cache simulator, showcasing performance optimization and memory and resource management.

Real vs Fake Face ML Classifier – Python

April - May 2024

- Built CNN & KNN models to classify real vs. fake faces; compared accuracy, speed and efficiency across approaches in a team of three.

Ludo board game – Python, Object Oriented Programming

Nov - Dec 2022

- Designed and developed OOP-based Ludo with code optimization, documentation, version control, and testing best practices.

Career Path Advice App (TiE)

May - Nov 2020

- Lead 4-person team to prototype career-path app; pitched and secured pre-seed funding from investors.

EXTRACURRICULARS

- **CMU Basketball Club** – former varsity national player, competitive sports experience demonstrating discipline & teamwork
- **CMU dance team (Sahara)** – Competitive performer, trained in choreography & stage presence.
- **Student Leadership Council Head** – Organized & led events with 2000+ participants, managing large cross functional teams